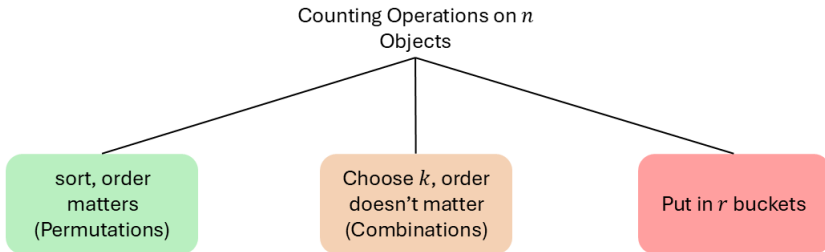

Probability and Statistic (for CS Students)

Lecture 4: Probability

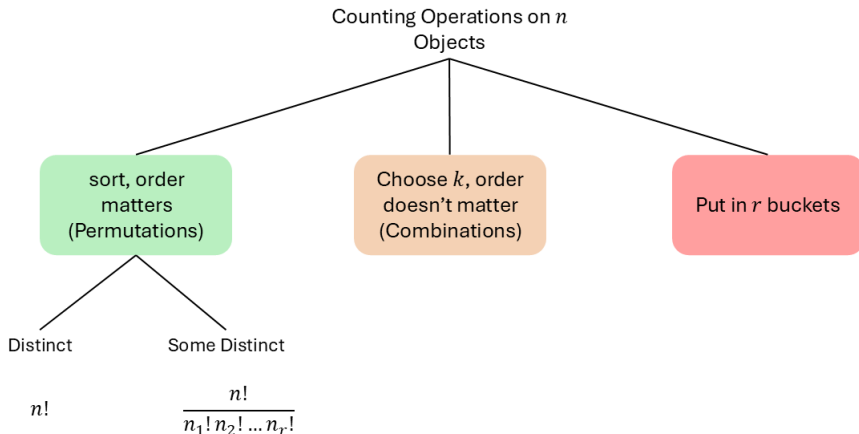
University of Mazandaran

Fall 1405

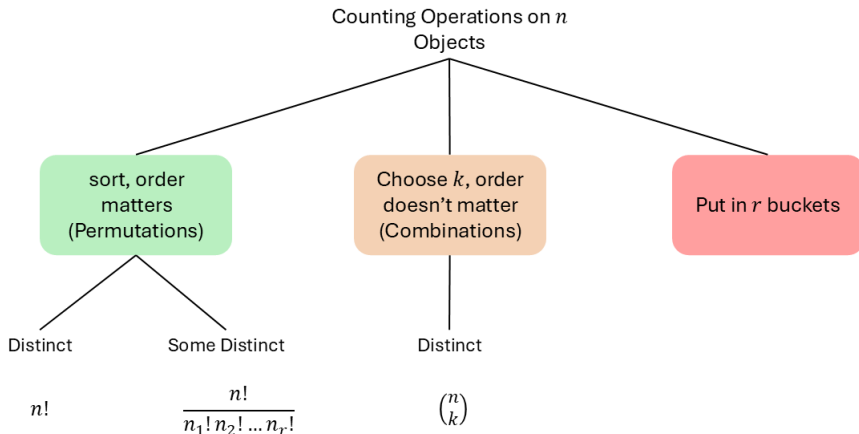
Counting & Combinatorics Recap



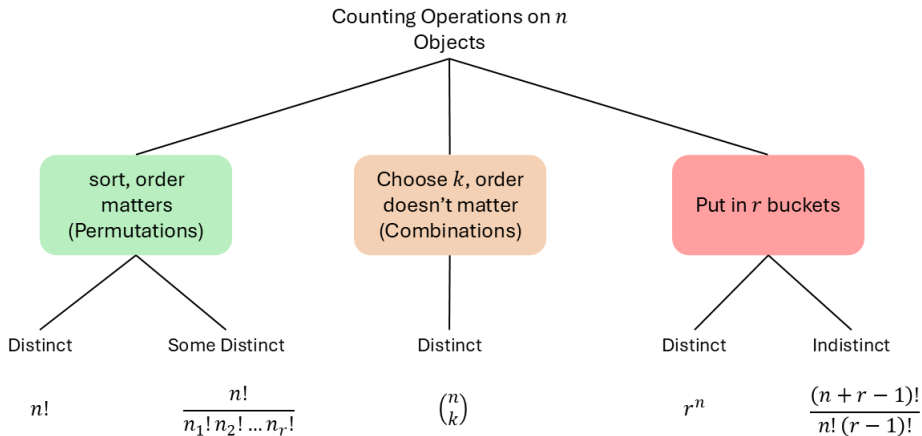
Counting & Combinatorics Recap



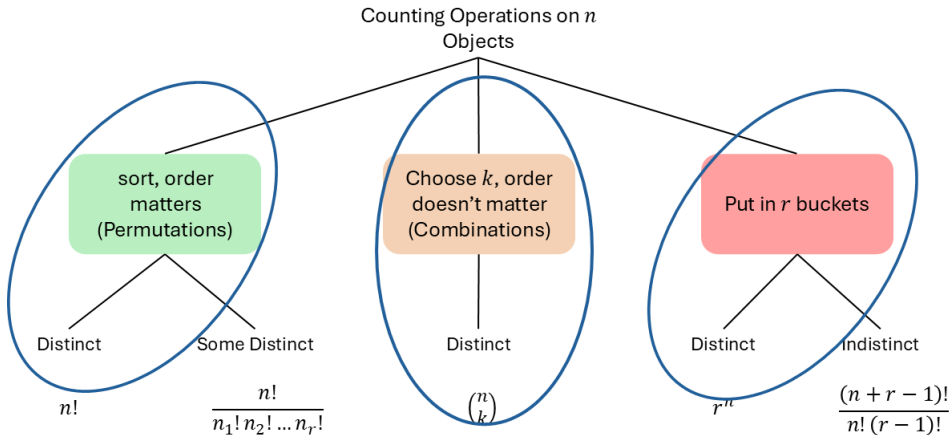
Counting & Combinatorics Recap



Counting & Combinatorics Recap



Counting & Combinatorics Recap



- ▶ Sample space, S , is set of all possible outcomes of an experiment
 - ▶ coin flip: $S = \{\text{Heads, Tails}\}$
 - ▶ flipping two coins: $S = \{[H, H], [H, T], [T, H], [T, T]\}$
 - ▶ roll of 6-sided die: $S = \{1, 2, 3, 4, 5, 6\}$
 - ▶ # emails in a day: $S = \{x | x \in \mathbb{Z}, x \geq 0\}$
 - ▶ screen time in day: $S = \{x | x \in \mathbb{R}, 0 \leq x \leq 24\}$

- ▶ Event, E , is some subset of S $\{E \subseteq S\}$
 - ▶ coin flip is heads: $E = \{\text{Heads}\}$
 - ▶ ≥ 1 head on 2 coin flips: $E = \{[H, H], [H, T], [T, H]\}$
 - ▶ roll of die is 3 or less: $E = \{1, 2, 3\}$
 - ▶ # emails in a day ≤ 20 : $E = \{x | x \in \mathbb{Z}, 0 \leq x \leq 20\}$
 - ▶ wasted day (≥ 5 screen time): $E = \{x | x \in \mathbb{R}, 5 \leq x \leq 24\}$

A number in $[0, 1]$ to which we ascribe meaning

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our belief that an event E might occur

A number in $[0, 1]$ to which we ascribe meaning



our belief that an event E might occur

A number in $[0, 1]$ which codes our belief that an event E might occur.

A number in $[0, 1]$ to which we ascribe meaning



our belief that an event E might occur

A number in $[0, 1]$ which codes our belief that an event E might occur.

Mathematically, we demonstrate it with $P(E)$

What is Probability

$$P(E) = \lim_{n \rightarrow \infty} \frac{\text{count}(E)}{n}$$

The event we care about

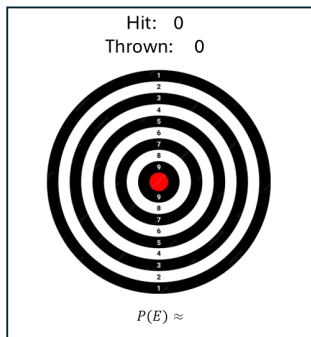
Out of (close to) infinite trials

How many times does it occur?

What is Probability

$$P(E) = \lim_{n \rightarrow \infty} \frac{\text{count}(E)}{n}$$

n is the number
of trials

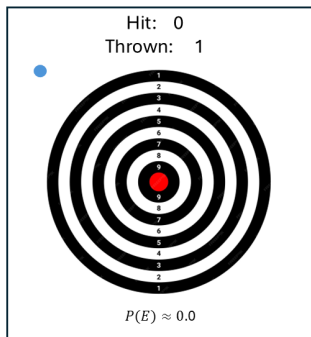


The "event" E is
that you hit the
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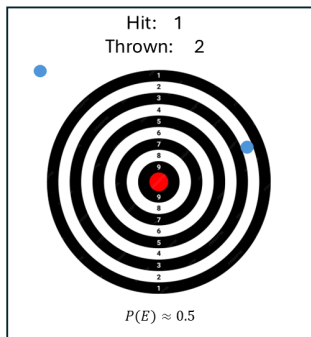


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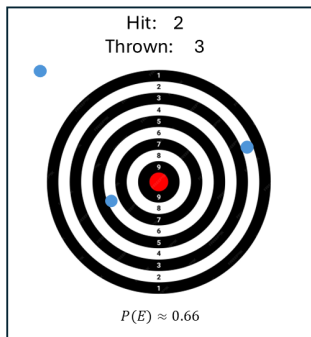


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$$P(E) = \lim_{n \rightarrow \infty} \frac{\text{count}(E)}{n}$$

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The "event" E is
that you hit the
target

What is Probability



Recall: S = all possible outcomes. E = the event.

- ▶ Axiom 1: $0 \leq P(E) \leq 1$
- ▶ Axiom 2: $P(S) = 1$
- ▶ Axiom 3: if events E and F are mutually exclusive:
 $P(E \cup F) = P(E) + P(F)$

Core Rules of Probability

Recall: S = all possible outcomes. E = the event.

- ▶ Axiom 1: $0 \leq P(E) \leq 1$
- ▶ Axiom 2: $P(S) = 1$
- ▶ Identity 3: $P(E') = 1 - P(E)$

Equally Likely Outcomes

Some sample spaces have **equally likely outcomes**.

- ▶ Coin flip: $S = \{\text{Head}, \text{Tails}\}$
- ▶ Flipping two coins: $S = \{[H, H], [H, T], [T, H], [T, T]\}$
- ▶ Roll of 6-sided die: $S = \{1, 2, 3, 4, 5, 6\}$

If we have equally likely outcomes, then $P(\text{each outcome}) = \frac{1}{|S|}$

Therefore $P(E) = \frac{\# \text{ of outcomes in } E}{\# \text{ of outcomes in } S} = \frac{|E|}{|S|}$ {by Axiom 3}

Not Everything is Equally Likely

Visa lottery application

- ▶ What is $P(\text{Win})$?



Not Everything is Equally Likely

Visa lottery application

► What is $P(\text{Win})$?

$S = \{\text{Lose}, \text{Win}\}$

$E = \text{Win}$

$$P(E) = \frac{|E|}{|S|} = \frac{1}{2}$$



Not Everything is Equally Likely

Roll two 6-sided dice. What is $P(\text{sum} = 7)$?

$$\left\{ \begin{array}{cccccc} [1,1] & [1,2] & [1,3] & [1,4] & [1,5] & [1,6] \\ [2,1] & [2,2] & [2,3] & [2,4] & [2,5] & [2,6] \\ [3,1] & [3,2] & [3,3] & [3,4] & [3,5] & [3,6] \\ [4,1] & [4,2] & [4,3] & [4,4] & [4,5] & [4,6] \\ [5,1] & [5,2] & [5,3] & [5,4] & [5,5] & [5,6] \\ [6,1] & [6,2] & [6,3] & [6,4] & [5,6] & [6,6] \end{array} \right\}$$

Not Everything is Equally Likely

Roll two 6-sided dice. What is $P(\text{sum} = 7)$?

Let E be the event that the sum is 7

$$\left\{ \begin{array}{cccccc} [1,1] & [1,2] & [1,3] & [1,4] & [1,5] & [1,6] \\ [2,1] & [2,2] & [2,3] & [2,4] & [2,5] & [2,6] \\ [3,1] & [3,2] & [3,3] & [3,4] & [3,5] & [3,6] \\ [4,1] & [4,2] & [4,3] & [4,4] & [4,5] & [4,6] \\ [5,1] & [5,2] & [5,3] & [5,4] & [5,5] & [5,6] \\ [6,1] & [6,2] & [6,3] & [6,4] & [5,6] & [6,6] \end{array} \right\}$$

$E =$ in blue

$P(E) =$

Not Everything is Equally Likely

Roll two 6-sided dice. What is $P(\text{sum} = 7)$?

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$$\left\{ \begin{array}{cccccc} [1,1] & [1,2] & [1,3] & [1,4] & [1,5] & [1,6] \\ [2,1] & [2,2] & [2,3] & [2,4] & [2,5] & [2,6] \\ [3,1] & [3,2] & [3,3] & [3,4] & [3,5] & [3,6] \\ [4,1] & [4,2] & [4,3] & [4,4] & [4,5] & [4,6] \\ [5,1] & [5,2] & [5,3] & [5,4] & [5,5] & [5,6] \\ [6,1] & [6,2] & [6,3] & [6,4] & [5,6] & [6,6] \end{array} \right\}$$

$E =$ in blue

$$P(E) = \frac{|E|}{|S|}$$

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$E =$ in blue

$$P(E) = \frac{|E|}{|S|} = \frac{6}{36} = .16\bar{6}$$

Python Implementation

```
import random
from tqdm import tqdm

N_TRIALS = 10000000 # getting close to infinity
TARGET_SUM = 7      # event

def main():
    n_events = 0
    for i in tqdm(range(N_TRIALS)):
        dice_total = run_experiment()
        if dice_total == TARGET_SUM:
            n_events += 1
    pr_e = n_events / N_TRIALS
    print(f'After {i} trials: ', 'P(E) = ', pr_e)

def run_experiment():
    d_1 = roll_dice()
    d_2 = roll_dice()
    return d_1 + d_2

def roll_dice():
    # this simulates rolling a die
    return random.choice([1,2,3,4,5,6])

if __name__ == '__main__':
    # this starts the program
    main()
```

Sum of Two Dice = 2?

Roll two 6-sided dice. What is $P(\text{sum} = 2)$?

Let E be the event that the sum is 2

$$\left\{ \begin{array}{cccccc} [1,1] & [1,2] & [1,3] & [1,4] & [1,5] & [1,6] \\ [2,1] & [2,2] & [2,3] & [2,4] & [2,5] & [2,6] \\ [3,1] & [3,2] & [3,3] & [3,4] & [3,5] & [3,6] \\ [4,1] & [4,2] & [4,3] & [4,4] & [4,5] & [4,6] \\ [5,1] & [5,2] & [5,3] & [5,4] & [5,5] & [5,6] \\ [6,1] & [6,2] & [6,3] & [6,4] & [5,6] & [6,6] \end{array} \right\}$$

$E =$ in red

$P(E) =$

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$E =$ in red

$$P(E) = \frac{|E|}{|S|} = \frac{1}{36} = 0.02\bar{7}$$

Other Ways to Make Sample Space

sum of two dice: three options for the sample space

Think of the die as

distinct

$[5, 5]$



Think of the die as

indistinct

$\{5, 5\}$

Just look at the sum

10

Other Ways to Make Sample Space

Roll two 6-side dice. What is probability the sum = 7?

Let S be the sum of two dice and E be the event that the sum is 7

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$P(E) = \frac{|E|}{|S|} = \frac{1}{11} = 0.0\bar{83}$$

Other Ways to Make Sample Space

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Think of the die as

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Other Ways to Make Sample Space

Roll two 6-sided dice. What is $P(\text{sum} = 7)$

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$E =$ in blue

$$P(E) = \frac{|E|}{|S|} = \frac{3}{20} = 0.15$$

Other Ways to Make Sample Space

sum of two dice: three options for the sample space

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$[5, 5]$



Think of the die as

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Just look at the sum

10



Cats and Dogs

- ▶ 4 cats and 3 dogs in a toy box. 3 drawn.
- ▶ what is $P(1 \text{ cat and } 2 \text{ dogs drawn})$



The Choice of Sample Space is Yours

	Distinct	Indistinct
Unordered		
Ordered		

The Choice of Sample Space is Yours

	Distinct	Indistinct
Unordered	$\{c_1, d_2, d_3\}$ $\{c_1, c_2, c_3\}$	
Ordered		

The Choice of Sample Space is Yours

	Distinct	Indistinct
Unordered	$\{c_1, d_2, d_3\}$ $\{c_1, c_2, c_3\}$	$\{2 \text{ cats}, 1 \text{ dog}\}$ $\{3 \text{ cats}\}$
Ordered		

The Choice of Sample Space is Yours

	Distinct	Indistinct
Unordered	$\{c_1, d_2, d_3\}$ $\{c_1, c_2, c_3\}$	$\{2 \text{ cats}, 1 \text{ dog}\}$ $\{3 \text{ cats}\}$
Ordered	$[c_1, d_2, d_3]$ $[c_1, c_2, c_3]$	

The Choice of Sample Space is Yours

	Distinct	Indistinct
Unordered	$\{c_1, d_2, d_3\}$ $\{c_1, c_2, c_3\}$	{2 cats, 1 dog} {3 cats}
Ordered	$[c_1, d_2, d_3]$ $[c_1, c_2, c_3]$	[cat, dog, cat] [cat, cat, cat]

The Choice of Sample Space is Yours

	Distinct	Indistinct
Unordered	$\{c_1, d_2, d_3\}$ $\{c_1, c_2, c_3\}$	{2 cats, 1 dog} {3 cats}
Ordered	$[c_1, d_2, d_3]$ $[c_1, c_2, c_3]$	[cat, dog, cat] [cat, cat, cat]

Which choice will lead to equally likely outcomes?

The Choice of Sample Space is Yours

	Distinct	Indistinct
Unordered	$\{c_1, d_2, d_3\}$ $\{c_1, c_2, c_3\}$	{2 cats, 1 dog} {3 cats}
Ordered	$[c_1, d_2, d_3]$ $[c_1, c_2, c_3]$	[cat, dog, cat] [cat, cat, cat]

Which choice will lead to equally likely outcomes?

Distinct

- ▶ 4 cats and 3 dogs. 3 drawn.
 - ▶ what is $P(1 \text{ cat and } 2 \text{ dogs drawn})$
- ▶ Unordered

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- ▶ 4 cats and 3 dogs. 3 drawn.
 - ▶ what is $P(1 \text{ cat and } 2 \text{ dogs drawn})$
- ▶ Unordered
 - ▶ $|S| = \binom{7}{3} = 35$
 - ▶ $|E| = \binom{4}{1} \binom{3}{2} = 12$
 - ▶ $P(1 \text{ cat and } 2 \text{ dogs drawn}) = \frac{12}{35}$

Distinct

- ▶ 4 cats and 3 dogs. 3 drawn.
 - ▶ what is $P(1 \text{ cat and } 2 \text{ dogs drawn})$
- ▶ Unordered
 - ▶ $|S| = \binom{7}{3} = 35$
 - ▶ $|E| = \binom{4}{1} \binom{3}{2} = 12$
 - ▶ $P(1 \text{ cat and } 2 \text{ dogs drawn}) = \frac{12}{35}$
- ▶ Ordered

Distinct

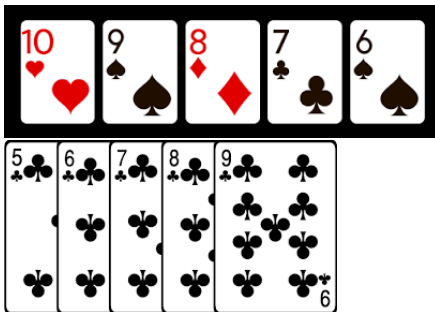
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 - ▶ what is $P(1 \text{ cat and } 2 \text{ dogs drawn})$
- ▶ Unordered
 - ▶ $|S| = \binom{7}{3} = 35$
 - ▶ $|E| = \binom{4}{1} \binom{3}{2} = 12$
 - ▶ $P(1 \text{ cat and } 2 \text{ dogs drawn}) = \frac{12}{35}$
- ▶ Ordered
 - ▶ pick 3 ordered items: $|S| = 7 \times 6 \times 5 = 210$
 - ▶ pick cat as either 1st, 2nd, or 3rd item:
 $|E| = 4 \times 3 \times 2 + 3 \times 4 \times 2 + 3 \times 2 \times 4 = 72$
 - ▶ $P(1 \text{ cat and } 2 \text{ dogs drawn}) = \frac{72}{210}$

Make indistinct items **distinct** to get
equally likely sample space outcomes

Challenging Question: Straight Poker Hand

Consider 5 card poker hand

- ▶ "Straight" is a 5 consecutive rank cards of any suit
- ▶ Order is not important
- ▶ What is $P(\text{straight})$



P.S.: This example is just for learning purposes.

P.S.: Homayun never played poker in his life!

Challenging Question: Straight Poker Hand

- ▶ Step 1: Sample space and its size
- ▶ Step 2: Event and its size

Challenging Question: Straight Poker Hand

- ▶ Step 1: Sample space and its size
 - ▶ select 5 cards out of 52 cards: $\binom{52}{5}$
- ▶ Step 2: Event and its size

Challenging Question: Straight Poker Hand

- ▶ Step 1: Sample space and its size
 - ▶ select 5 cards out of 52 cards: $\binom{52}{5}$
- ▶ Step 2: Event and its size
 - ▶ straight hand: $\binom{10}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1}$

Challenging Question: Straight Poker Hand

- ▶ Step 1: Sample space and its size
 - ▶ select 5 cards out of 52 cards: $\binom{52}{5}$
- ▶ Step 2: Event and its size
 - ▶ straight hand: $\binom{10}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1}$
- ▶ $P(\text{straight}) = \frac{|E|}{|S|} = \frac{\binom{10}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1} \cdot \binom{4}{1}}{\binom{52}{5}}$

key: When approaching an "equally likely probability" problem, start by defining sample space and event space.

Final Example

n chips are manufactured in a company and (somehow!) we know 1 chip is defective. If we select k chips randomly from the n chips, what is the probability that the defective chip is also among them?

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n chips are manufactured in a company and (somehow!) we know 1 chip is defective. If we select k chips randomly from the n chips, what is the probability that the defective chip is also among them?

$$|S| = \binom{n}{k}$$

$$|E| = \binom{1}{1} \binom{n-1}{k-1}$$

$$p = \frac{|E|}{|S|} = \frac{\binom{1}{1} \binom{n-1}{k-1}}{\binom{n}{k}} = \frac{k}{n}$$